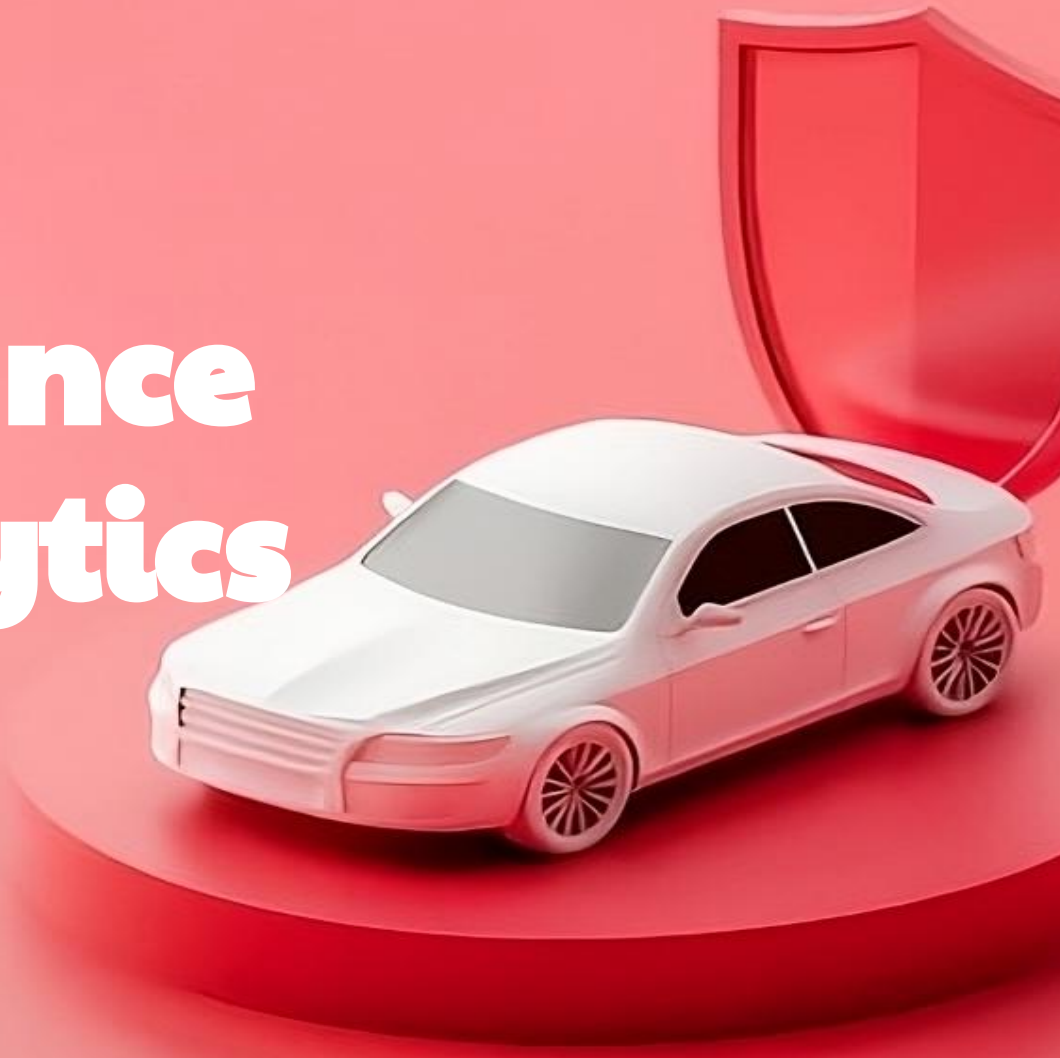
Car Insurance Loss Analytics

DSO 530 Group Project



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Project Overview

Industry Context

One of the main challenges in the insurance industry is setting the **right premium** for policyholders. **Mispricing** can cause **adverse selection** leading to **financial loss** .

Problem Statement

Insurers need to **accurately predict** expected **losses** and **adjust premiums** accordingly. Accurate losses prediction will help improve premium **pricing strategies** , reduce the risk of adverse selection, and enhance **overall portfolio profitability** .

Project Goal

Build **predictive models** to help insurance companies **set fair and profitable premiums**

Importance & Motivation

Reduce underwriting risk and support portfolio health

Promote fairness and efficiency in pricing using **Tweedie-based ML models**

Stakeholders

Policyholders
Seek transparent and justifiable premiums

Policymakers
Ensure market fairness and prevent discrimination

Insurance Companies
Need to maintain profitability

Task 1

Regression

Predict: how big the claim is

LC (Loss Cost per Exposure Unit) = X.15 / X.16

- Measure average cost per claim for a policyholder

HALC (Historically Adjusted Loss Cost) = LC × X.18

- Adjust LC to reflect how frequently claims occur historically

Business Perspective:

Determine **financial impact** when that risk materializes

Task 2

Classification

Predict: the probability of any claim

Claim Status (CS) :

- 1 = claim made
- 0 = no claim

Business Perspective:

Assess **risk of any claim**

Dataset Overview

Training Data

- 37,451 policy records
- 28 variables (X.1–X.28)

Feature Types

- **Numerical** : X.8–X.12, X.14–X.18, X.22–X.26, X.28
- **Categorical:** X.7, X.13, X.19–X.21, X.27
- **Datetime:** X.2–X.6

Test Data

- 15,787 policy records
- Without target values (X.15–X.18)

Datetime Conversion

- Convert **policy start date** , **renewal dates** , **date of birth** , and **license issuance date** to datetime.

Feature Engineering

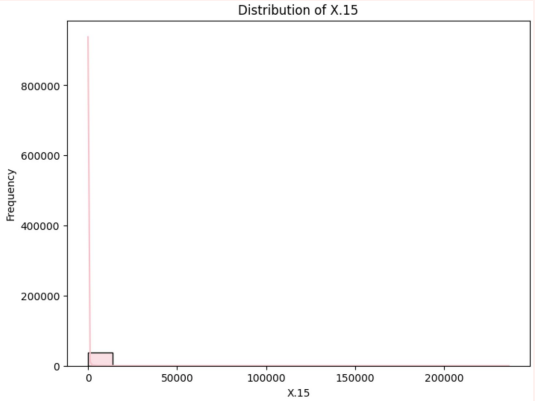
- **age** = Age of the policyholder in years
- **policy_duration** = Days between policy start and last renewal
- **driver_experience** = Years since driver's license was issued
- **Vehicle_age** = Age of vehicle as of 2019

Columns Dropped

- **X.1**: A unique identifier for each policy, which doesn't provide much useful information and could introduce unwanted noise.
- **X.2–X.6 and X.22:** We already extracted useful components using feature engineering.

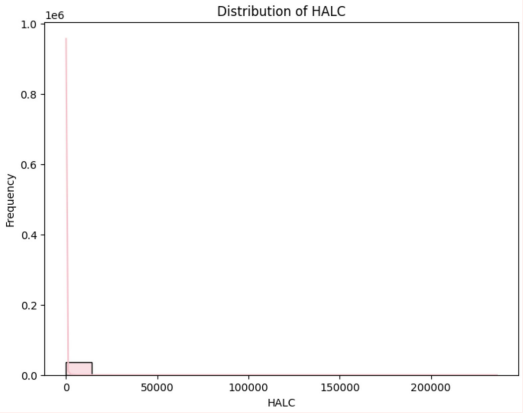
Distributions of Numerical Features (Used for LC, HALC, CS Prediction)

X.15–Total Claim Cost



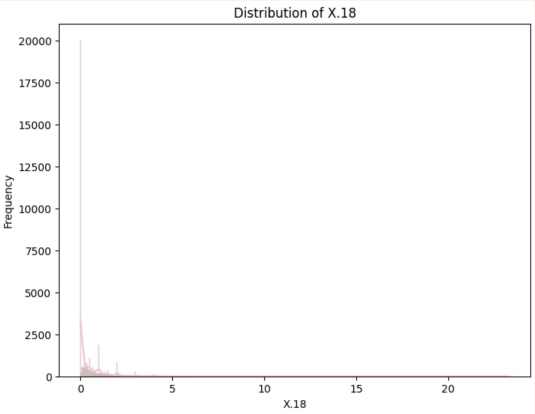
- Highly **right-skewed** with many **zeros** – most policyholders didn't file a claim.
- Few records show **extreme claim costs**, highlighting **rare but large incidents**.

X.15 / X.16–Loss Cost



- Extreme **spike at zero** and long tail – many low-cost claims, few expensive ones.
- Outliers with **very high per-claim costs**, possibly from rare high-payout policies.

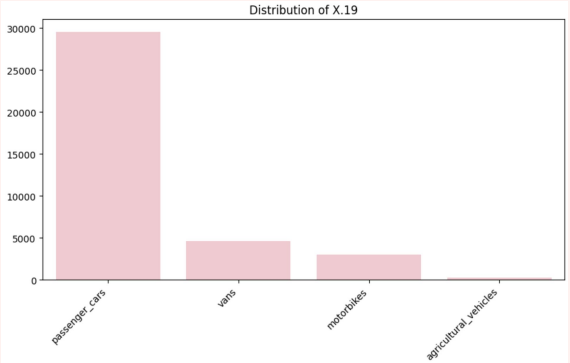
X.18–Claim Frequency



- **Heavily right-skewed**, with most values close to zero.
- Claims occur **infrequently** across most policies.

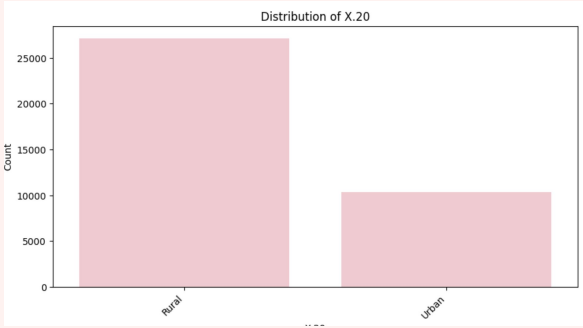
Distributions of Categorical Features

X.19–Vehicle Type



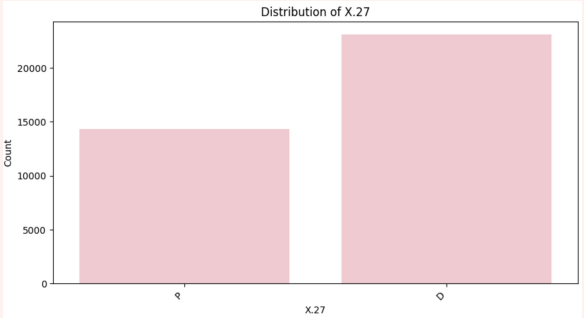
- **Passenger cars** make up the vast majority of insured vehicles.
- **Vans** and **motorbikes** represent much smaller segments.
- Very few **agricultural vehicles** , indicating they may have limited impact on the model.

X.20–Region Type



- The dataset is **dominated by rural customers** , with urban policyholders being a smaller group.

X.27–Fuel Type



- Majority of vehicles run on **diesel** , but **petrol** still forms a significant portion.
- Class balance is relatively better than other categorical features.

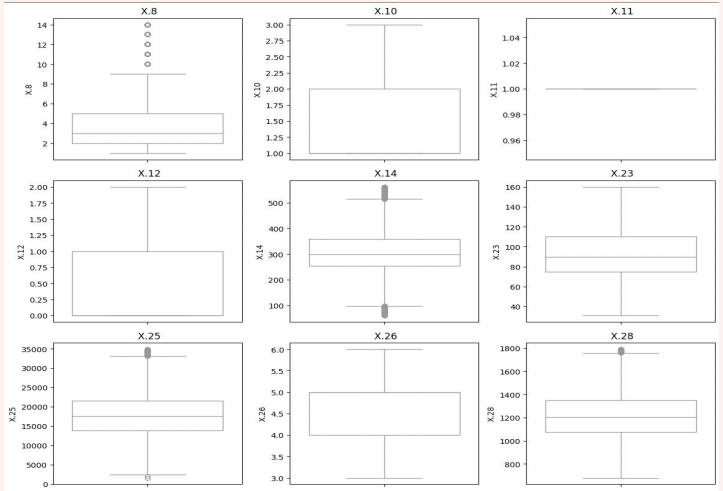
Handling Outliers

Regression

Thresholds Applied

Outliers beyond 1.5 × IQR from the 1st and 3rd quartiles

~36% of rows were removed in total



Classification

Thresholds Applied

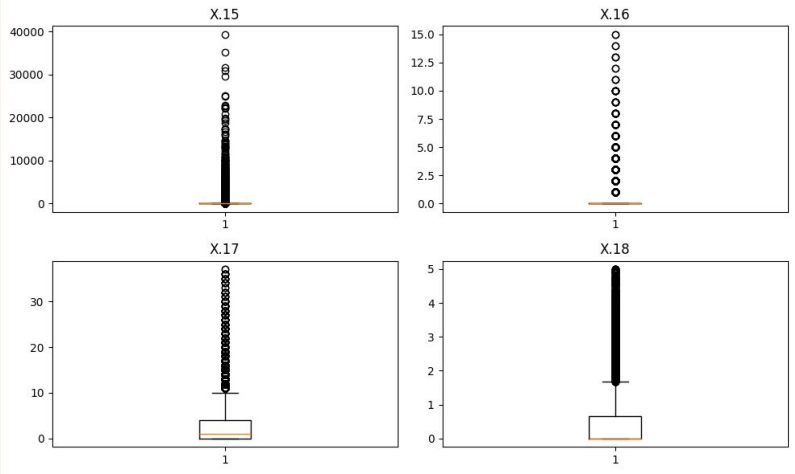
X.15 > 40000

X.16 > 15

X.17 > 40

X.18 > 10

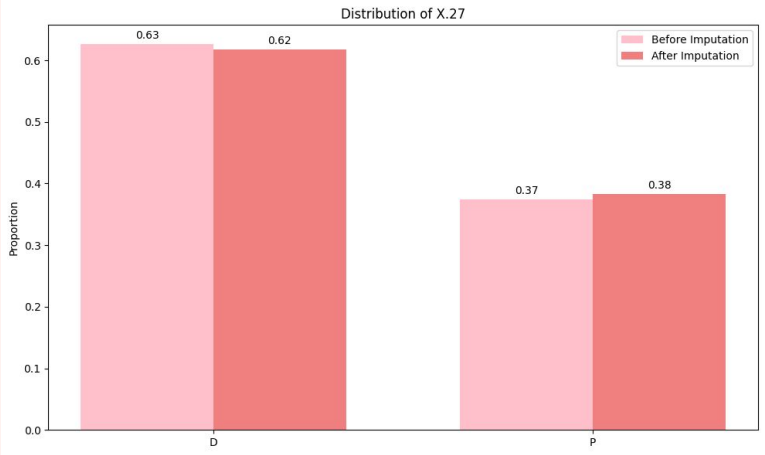
Less than 0.1% of rows were removed in total



Handling Missing Values in X.27 (Fuel Type)

Approach

- Imputed using supervised classification models
  - Features used: X.19, X.23, X.24, X.25, X.28
- Final imputed values were integrated back into the training dataset



Comparison

Model	Decision Tree / Random Forest	XGBoost	LightGBM
Accuracy	99.76%	96.31%	94.7%
Recall (D / P)	1.00 / 1.00	0.98 / 0.94	0.97 / 0.91
Observation	Overfit to training set, poor recall on minority class P	Best balance between performance and generalization; strong recall on both classes, especially minority P	Slight underfitting, missed both classes more often

Features Used & Correlation Heatmap

Columns with correlation > |0.8|

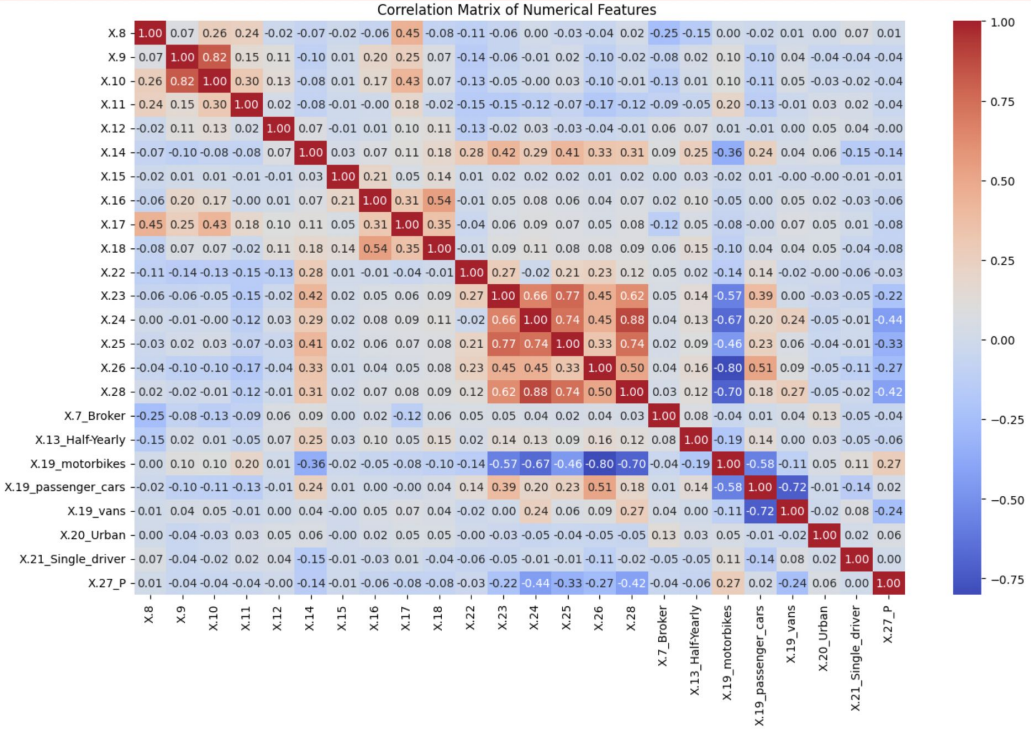
- 'X.9' and 'X.10' = 0.82
- 'X.24' and 'X.28' = 0.88
- 'X.26' and 'X.19_motorbikes' = -0.8

Columns dropped

Drop **'X.9', 'X.24', 'X.19_motorbikes'** columns apart from date and year columns

Final features used

- **Numerical:** 'X.8', 'X.10', 'X.11', 'X.12', 'X.14', 'X.23', 'X.25', 'X.26', 'X.28'
- **Categorical:** 'X.7_Broker', 'X.13_Half-Yearly', 'X.19_passenger_cars', 'X.19_vans', 'X.20_Urban', 'X.21_Single_driver', 'X.27_P'
- **Prediction:** 'LC', 'HALC', 'CS'



Regression *

Prediction Target: LC & HALC
Evaluation Metric: RMSE

Model Type	Model	LC	HALC
Linear	Tweedie Regression	573.67	1078.13
Non-Linear	LightGBM	571.79	1072.38
	XGBoost	574.04	932.68
	Neural Network	**458.92**	**718.54**

*All models trained using Tweedie Loss Function

Classification

Prediction Target: CS
Evaluation Metric: ROC-AUC

Model Type	Model	ROC-AUC Score
Linear	Logistic Regression	0.74
Non-Linear	**XGBoost**	**0.85**
	Random Forest	0.82
	Neural Network	0.71

Business Understandings

Business Perspective

- **Prediction accuracy** is more critical for setting proper premiums.
- Accurate risk prediction ensures **profitability** and protects **against adverse selection** .
- The cost of inaccurate predictions outweighs the need for full model transparency.

Key Challenges

- **Highly skewed and zero-inflated data:** Rare, costly claims challenge prediction.
- **Complexity vs. Interpretability:** Models must balance accuracy and transparency.
- **Generalization to new customers:** Models must adapt to unseen policyholders.

Variable Selection

- Driver characteristics
- Vehicle attributes
- Policy details
- Location information

Applicability

Car insurance model cannot be directly used for life insurance.

- Different risk objects (vehicle vs human life)
- Different predictive features

THANK YOU!